

Welcome & Overview

Edinburgh Summer School on Generative AI for Extremes

Miguel de Carvalho

University of Edinburgh
Chair of the Event

8–11 September 2026
JCMB, Edinburgh



THE UNIVERSITY
of EDINBURGH



The GAMEX Team

A Growing Network across Una Europa



Luca Trapin, Miguel de Carvalho, Massimo Ventrucchi, Ivo Bonfanti, Johnny Lee, Giorgia Ramponi,
Till Freihaut, Brendan Murphy, Luis Gutierrez Inostroza and Clemente Ferrer



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Zürich



University College Dublin
An Colaiste Ollscoile, Baile Átha Cliath



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DE CHILE

Examples of GAMEX Outputs

Training



**Pontificia
Universidad
Católica de Chile**

Examples of GAMEX Outputs

Research, Editorial Work and Shared Resources



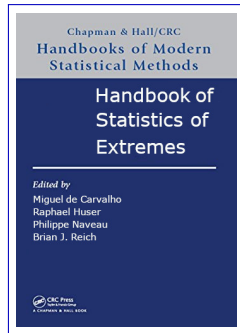
Special Issue

Bridging Heavy Tails and
Artificial Intelligence



New Research

Neural Modelling of Cascading
Extremes



Shared Resource

Handbook of Statistics of
Extremes

GLEN

Glasgow–Edinburgh Extremes Network



First GLEN workshop, Glasgow



GLEN

Glasgow–Edinburgh
Extremes Network

Now Expanding

Lancaster, London and
Newcastle

GAME 2025 and GAME 2026

From Edinburgh to Bologna



GAME 2025

Edinburgh



GAME 2026

Bologna

GAME 2025 was the first worldwide workshop on generative AI modelling for extreme events.

The University of Edinburgh

A World-Leading Tradition in AI, Data Science and Statistics



Geoffrey Hinton

PhD in AI, 1978

Often called the godfather of AI

2024 Nobel Prize in Physics



AI and Data Science

Edinburgh has led European AI research since 1963.

Alumni include **Geoffrey Hinton** and **Christopher Bishop**.

Statistics

Thomas Bayes studied here from 1719 to 1722.

More recent alumni include **Nick Jewell**, former co-editor of JASA.

Programme

Day 1: Tuesday, 8 September

Day 1: Tuesday

8 September

Theme: *Foundations and Preparations for Generative Modelling*

Moderator: Luca Trapin

Time	Session	Lead
09:00–09:25	Registration & Welcome Coffee	Organisers
09:25–09:30	<i>Welcome & Overview</i>	Chair
09:30–11:00	<i>Introducing Extreme Value Theory</i>	Miguel de Carvalho
11:00–11:30	Coffee break	
11:30–12:30	<i>On Neural Generative Models</i>	Miguel de Carvalho
12:30–14:00	Lunch	
14:00–15:00	<i>Probabilistic Aspects of Diffusions</i>	Sotirios Sabanis
15:00–15:30	Coffee break	
15:30–17:00	<i>Practical: Neural Generative Models in Practice</i>	Johnny Lee

† Lectures: Lecture Theatre B; practical sessions: Teaching Studio 3217.

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Today: no generative extremes yet, only preparation in extremes and generative modelling.

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Programme

Day 2: Wednesday, 9 September

Day 2: Wednesday

9 September

Theme: *Generative Modelling for Extremes and Risk*

Moderator: Miguel de Carvalho

Time	Session	Lead
09:30–10:30	<i>GAN-based Generative Approaches for Extremes</i>	Stéphane Girard (online)
10:30–11:00	Coffee break	
11:00–12:00	<i>GAN-based Modelling for Operational Risk</i>	Luca Trapin
12:00–13:30	Lunch	
13:30–15:30	<i>Practical: Generative Modelling for Extremes and Risk</i>	Jordan Richards
15:30–16:00	Coffee break	
16:00–16:10	Group photo	
16:10–17:10	<i>Open Practice & Office Hours</i>	Instructors

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Programme

Day 3: Thursday, 10 September

Day 3: Thursday

10 September

Theme: *Multivariate Extremes, Cascades, and Simulation*

Moderator: Giorgia Ramponi

Time	Session	Lead
09:30–10:30	<i>Cascades, Multivariate Extremes, and Transformers</i>	Miguel de Carvalho
10:30–11:00	Coffee break	
11:00–12:00	<i>Multivariate Angular Simulation</i>	Emma Simpson
12:00–13:30	Lunch	
13:30–15:30	<i>Practical: Transformer-Based Approaches for Extremes</i>	Clemente Ferrer
15:30–16:00	Coffee break	
16:00–17:00	<i>Open Practice & Office Hours</i>	Instructors

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16:00–17:00	Open Practice & Office Hours	Instructors

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Programme

Day 4: Friday, 11 September

Day 4: Friday

11 September

Theme: *Other Frontier Topics*

Moderator: Emma Simpson

Time	Session	Lead
09:30–10:30	<i>Reinforcement Learning</i>	Giorgia Ramponi
10:30–11:00	Coffee break	
11:00–12:00	<i>Pre-trained Extreme Value Modelling</i>	Jordan Richards
12:00–12:10	<i>Final Announcements</i>	—
12:10–12:20	<i>Closing Remarks</i>	Chair
12:20–13:50	Lunch	

† Lectures: Lecture Theatre B; practical sessions: Teaching Studio 3217.

Partners & Organisers

Thank You

Partners



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THE UNIVERSITY of EDINBURGH
Generative AI Laboratory



THE UNIVERSITY of EDINBURGH
Edinburgh Centre for Financial Innovations



THE UNIVERSITY of EDINBURGH
Centre for Investing Innovation

In partnership with
aabrdr



LEVERHULME
TRUST



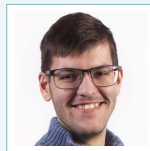
European Centres in Advanced Statistics



Organisers



Johnny Lee



Jordan Richards



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Welcome to Edinburgh!

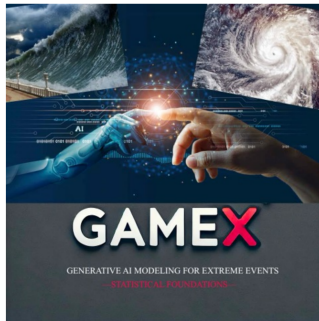
8–11 September 2026

Lecture Theatre B for lectures
Teaching Studio 3217 for practical
sessions

Edinburgh Summer School on Generative AI for Extremes



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Programme—updated 6 September 2026
8–11 September 2026

<https://games-networks.github.io/school/>

GL'EN (Glasgow–Lancaster–London–Edinburgh–Newcastle) Extremes Network

GAMEX Project Members: Luca Tadini • Miguel de Caceres • Massimo Vizzari • Ivo Bonfanti • Jinhua Lee • Georgia Roussou
Till Bracke • Brendan Murphy • Luis Guillermo Ibarra • Clemente Pardo